Clause projectors implement $\mathrm{e}^{-i\gamma H_c}$ without densifying H_c or matrix-exponentiating 2^n

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Abstract

For TEB 4-SAT with mutually commuting Fock clause projectors C_c, the product of local gadgets $U_c(\gamma) = I + (e^{-i\gamma} - 1)C_c$ equals $\exp(-i\gamma H_c)$ with $H_c = \sum_c C_c$ *exactly*. Each gadget is applied sparsely (Fock-local / SNAP mask), so one never builds the densified diagonal $E \in \mathbb{C}^{2^n}$ nor a $2^n \times 2^n$ matrix exponential. Conjugation by $W = H^{\otimes 3}$ only changes frame; the identity still holds in the computational (Fock) frame. This note records the algebra and the complexity contrast with Arm C densified diag(E).

1 Setup

Fix TEB 4-SAT on n bits packed into cavities of Fock dimension 8 (3 bits per cavity). For each clause c let C_c be the *Fock clause projector*: the orthogonal projector onto the subspace of computational basis states that *falsify* clause c. Then

$$C_c^\dagger = C_c, \qquad C_c^2 = C_c, \tag{1}$$

and, because every C_c is diagonal in the same Fock / computational basis,

$$[C_c, C_{c'}] = 0 \qquad \text{for all clauses } c, c'. \tag{2}$$

Define the cost Hamiltonian

$$H_c \;:=\; \sum_c C_c. \tag{3}$$

On computational basis state $|x\rangle$, $H_c|x\rangle = E(x)|x\rangle$ where $E(x)$ is the number of unsatisfied clauses (the densified energy used by Arm C).

2 Single-clause gadget

Lemma 1 (Projector exponential)**.** *Let P be an orthogonal projector and $\gamma \in \mathbb{R}$. Then*

$$U(\gamma) \;:=\; I + (e^{-i\gamma} - 1)\, P \;=\; \exp(-i\gamma P). \tag{4}$$

Proof. Since $P^2 = P$, the spectral theorem gives $P = \sum_j |\psi_j\rangle\langle\psi_j|$ on its support with eigenvalues in $\{0, 1\}$. Hence

$$\begin{aligned} \exp(-i\gamma P) &= \sum_{k=0}^{\infty} \frac{(-i\gamma)^k}{k!}\, P^k = I + \sum_{k=1}^{\infty} \frac{(-i\gamma)^k}{k!}\, P \\ &= I + \left(e^{-i\gamma} - 1\right) P, \end{aligned} \tag{5}$$

because $P^k = P$ for all $k \geq 1$. Equivalently, on $\ker P$ the operator acts as I, and on $\operatorname{ran} P$ it acts as $e^{-i\gamma} I$. □

Corollary 1. *For each clause projector C_c,*

$$U_c(\gamma) \;:=\; I + (e^{-i\gamma} - 1)\, C_c \;=\; \exp(-i\gamma C_c). \tag{6}$$

3 Product over clauses

Theorem 1 (Projector product = $\mathrm{e}^{-i\gamma H_c}$)**.** *If $\{C_c\}$ are mutually commuting orthogonal projectors, then*

$$\prod_c U_c(\gamma) \;=\; \prod_c \exp(-i\gamma C_c) \;=\; \exp\Big(-i\gamma \sum_c C_c\Big) \;=\; \exp(-i\gamma H_c). \tag{7}$$

The product may be taken in any order.

Proof. By Lemma 1, $U_c(\gamma) = \exp(-i\gamma C_c)$. Because $[C_c, C_{c'}] = 0$ for all c, c', the family $\{-i\gamma C_c\}$ also pairwise-commutes, so the Baker–Campbell–Hausdorff formula truncates:

$$\prod_c \exp(-i\gamma C_c) \;=\; \exp\Big(\sum_c (-i\gamma C_c)\Big) \;=\; \exp(-i\gamma H_c). \tag{8}$$

(Equivalently: all C_c are simultaneously diagonalizable; on a common eigenbasis vector with eigenvalues $p_c \in \{0, 1\}$, both sides multiply the amplitude by $\prod_c e^{-i\gamma p_c} = e^{-i\gamma \sum_c p_c}$.) □

Remark 1 (Honesty)**.** Theorem 1 shows that the projector "cost unitary" is *the same physics* as Arm C's $\mathrm{e}^{-i\gamma H_c}$. The gain is only algorithmic: one never densifies E of length 2^n. It is *not* a different U_c that secretly avoids H_c.

4 Complexity: sparse gadgets vs densified $\mathrm{diag}(E)$

Densified path (Arm C). Enumerate all 2^n bitstrings, compute $E(x) = \#\{\text{unsat clauses}\}$, store the vector $E \in \mathbb{R}^{2^n}$, then apply $\psi \mapsto e^{-i\gamma E} \odot \psi$ (pointwise) each layer. Memory and build cost scale as $\Theta(2^n)$; for $n = 18$ one E is already $\sim 1\,\mathrm{MB}$ float32, and $n = 24$ would be $\sim 64\,\mathrm{MB}$ per E.

Projector / SNAP-library path. Never form H_c or U as a dense $2^n \times 2^n$ matrix. Apply each clause gadget as a Fock-local update: for a span-1 clause the support is an 8-level SNAP phase mask; for span-2 a joint 8×8 SNAP; in all cases the per-clause work is $O(\mathrm{poly}(n))$ (constant in Fock dimension for fixed packing). With m clauses the layer costs

$$O\big(m \cdot \mathrm{poly}(n)\big) \tag{9}$$

(or $O(m)$ per amplitude if amplitudes are already materialized). The offline library of 26 local masks is compiled once ($\sim 1\,\mathrm{KB}$) and reused; only the shared angles γ_ℓ change per layer. By Theorem 1 the composed unitary remains exactly $\mathrm{e}^{-i\gamma H_c}$.

Proposition 1 (No densify, no matrix-exp of 2^n)**.** *Implementing $\prod_c U_c(\gamma)$ via sparse / Fock-local gadgets implements $\mathrm{e}^{-i\gamma H_c}$ without (i) allocating $E \in \mathbb{C}^{2^n}$, and without (ii) forming or exponentiating a dense $2^n \times 2^n$ matrix.*

5 $W = H^{\otimes 3}$ conjugation

Let $W = H^{\otimes 3}$ act on one cavity's three qubits (Hadamard on each bit). Cost implemented in the W-frame is

$$W^\dagger \, \mathrm{e}^{-i\gamma H_c} \, W \;=\; \exp\!\big(-i\gamma \, W^\dagger H_c W\big). \tag{10}$$

This is a *change of frame*: the spectrum of the generator is unchanged, and sandwich checks already verify $W^\dagger Z W = X$ and Fock$\leftrightarrow W$ fidelity 1 for the compiled SNAP masks. In the computational / Fock frame the product identity of Theorem 1 continues to hold; W does not break $\prod_c U_c = \mathrm{e}^{-i\gamma H_c}$.

6 What this does *not* claim

- It does *not* claim a U_c algebraically inequivalent to $\mathrm{e}^{-i\gamma H_c}$. Non-$H_c$ probes (surrogate, non-commuting XZ, wrong-graph, adaptive feedback, nonlinear clause phases) all die on n15 screens versus trunc_Hc / Arm C.
- It does *not* remove instance dependence: clauses still enter via $\{C_c\}$.
- Simulator wall-clock at small n may favor NumPy $\exp(-i\gamma E)$ over a pure-Python clause loop; that is an implementation artifact, not a complexity rebuttal (see `docs/COMPILE_OFFLINE.md`).

7 Summary

Claim	Status	Note
$U_c(\gamma) = I + (e^{-i\gamma} - 1)C_c = \mathrm{e}^{-i\gamma C_c}$	Proven	Lemma 1
$\prod_c U_c(\gamma) = \mathrm{e}^{-i\gamma H_c}$	Proven	Thm. 1, needs $[C_c, C_{c'}] = 0$
Sparse apply, no densify $E \in \mathbb{C}^{2^n}$	Complexity	Prop. 1
$W = H^{\otimes 3}$ keeps identity in Fock frame	Frame only	Sec. 5
Same physics as Arm C densified diag(E)	Honest	Remark after Thm. 1

One-liner. Clause projectors are an *exp-avoiding implementation* of $\mathrm{e}^{-i\gamma H_c}$—a complexity win over densification, not different cost physics.